Drug Quantification Framework

A 4-Level Multi-Dimensional Drug Profiling Framework

Proof of Concept - NSAID Class

July 2026

Built on MedQuery PubMed RAG (27.7M abstracts) + Inxight FRDB + DrugBank + Cochrane reviews

4 PoC drugs: ibuprofen (baseline), diclofenac (complexity), celecoxib (selectivity), paracetamol (stress test)

Table of Contents

1. Summary

Right now if you compare two analgesics you get a number - NNT, NNH, effect size. That number collapses mechanism, safety, pharmacokinetics, everything into one dimension. But paracetamol and ibuprofen don't share a mechanism, don't share a risk profile, and the "right" choice depends on whether your patient is bleeding, has heart disease, or just needs a tooth pulled. You can't answer that with a single number. The framework keeps all four dimensions separate and lets the question decide the weight.

This proof of concept profiles four analgesics - ibuprofen, diclofenac, celecoxib, and paracetamol - across four levels: molecular binding (L1), pharmacokinetics (L2), systems response (L3), and clinical outcomes (L4).

DQF is related to, but distinct from, existing multi-criteria decision analysis (MCDA) frameworks used in health technology assessment (e.g., EUnetHTA, PrOACT-URL, EMA benefit-risk models). Standard MCDA frameworks assign weights to criteria and produce a single utility score - which collapses dimensions in the same way NNT does. DQF instead preserves all four dimensions as independent axes and explicitly refuses to produce a ranking. The causal chain (L1->L3->L4, modulated by L2) is also unique: existing drug-information resources (DrugBank, ChEMBL, DrugCentral, Inxight FRDB, PharmGKB) each cover one or two levels but none integrate L1 through L4 into a single causal profile. DQF is not a scoring system. It is a drug fingerprint representation.

Drug	Role	Why Selected
Ibuprofen	Baseline reference	Most-studied NSAID, excellent data density
Diclofenac	Complexity	P2X3 off-target, unique biliary PK
Celecoxib	Selectivity	COX-2 selective, coxib paradox
Paracetamol	Stress test	Prodrug (AM404), multi-target, not NSAID

2. Framework Architecture

The framework profiles drugs across four independent levels. Each level is populated from structured databases, literature mining via PubMed RAG, and systematic reviews. Levels are connected by a causal chain (L1->L3->L4) modulated by L2 pharmacokinetics.

Level	What It Captures	Source	Key Metrics
L1 - Binding	Drug-receptor interaction	PDSP + PubMed RAG	Ki, Kd, IC50
L2 - PK	ADME properties	Inxight FRDB, DrugBank	F%, t1/2, Vd
L3 - Systems	Tissue/pathway biology	Literature (RAG)	COX dynamics, tissue t1/2
L4 - Clinical	Efficacy + safety	Cochrane, Oxford	NNT, NNH

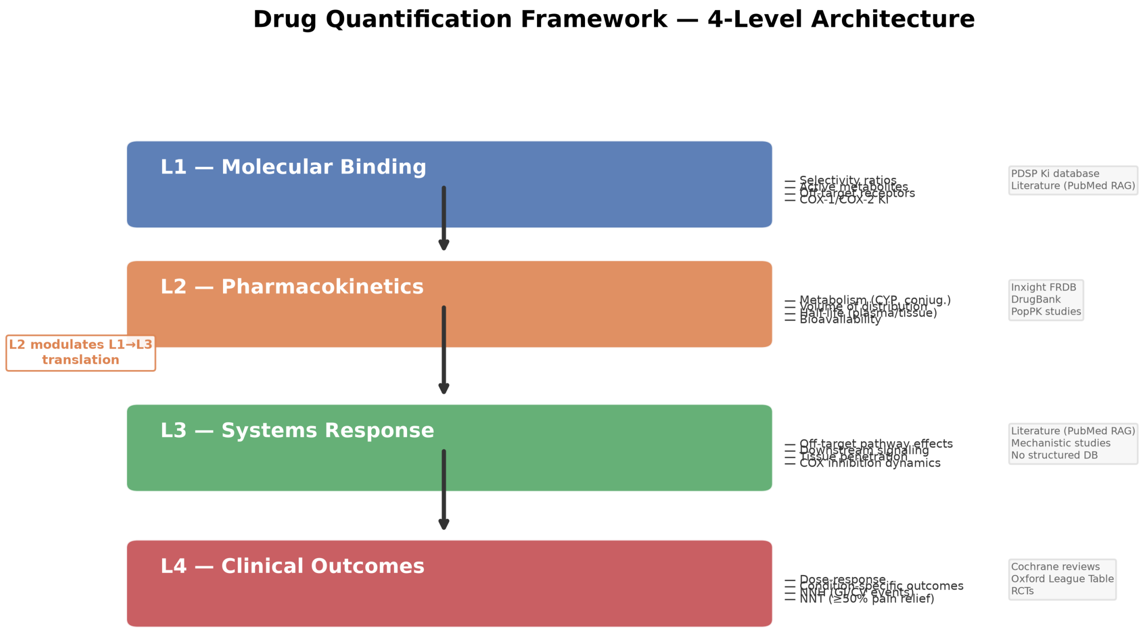

Figure 1. DQF 4-level architecture showing the flow from molecular binding through pharmacokinetics and systems response to clinical outcomes. L2 modulates the translation from L1 to L3.

3. L1 - Molecular Binding

Binding affinity profiles reveal that no two drugs in this set share a target profile. Diclofenac has unique P2X3 antagonism (though at micromolar concentrations above typical plasma levels). Ibuprofen uniquely inhibits ASIC1a allosterically. Paracetamol acts entirely through its metabolite AM404, which is multi-target. Celecoxib is distinguished by COX-2 selectivity (~30:1 vs COX-1).

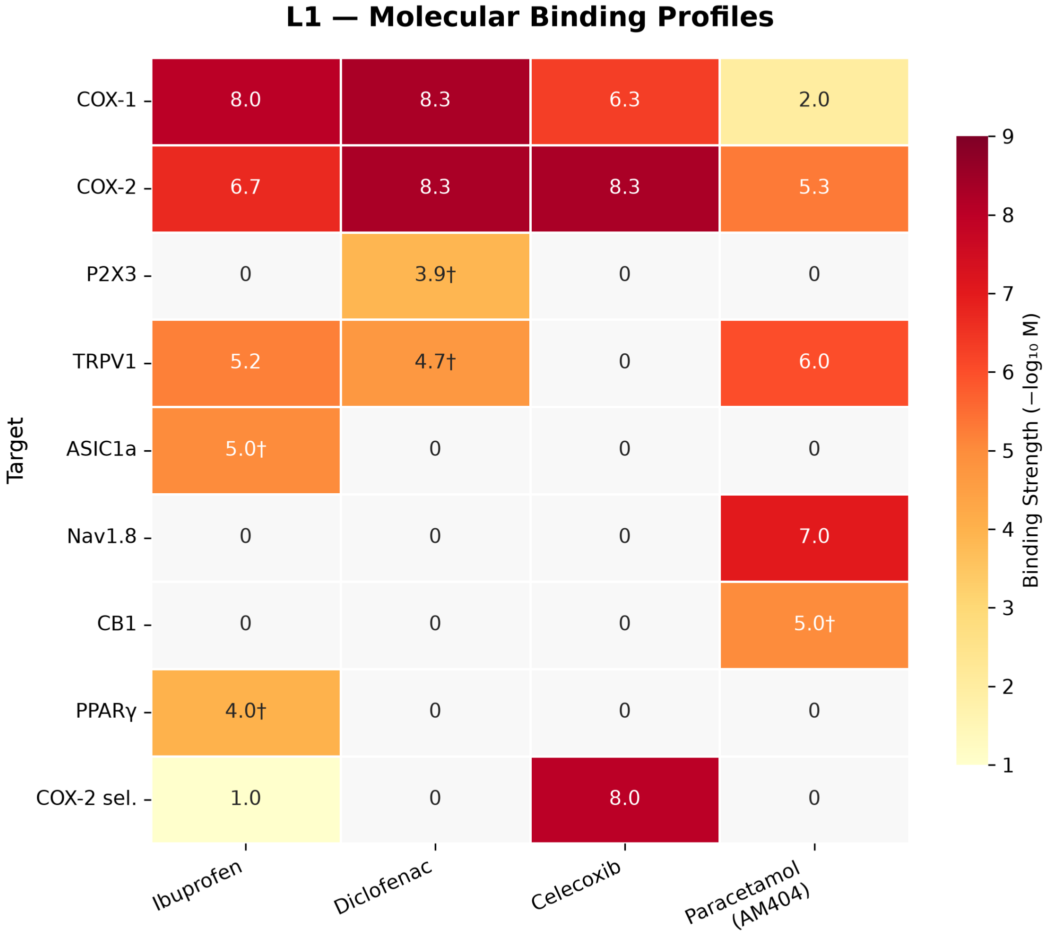

Figure 2. L1 binding profiles. Values are -log10 of IC50/Ki in molar: higher = stronger binding. Zero indicates no measurable activity.

Target	Ibuprofen	Diclofenac	Celecoxib	Paracetamol
COX-1	~10 nM	~5 nM	~500 nM	>100 uM
COX-2	~200 nM	~5 nM	~5 nM	~5 uM (weak)
COX-2 sel.	~0.05	~1	~30:1	Neglig.
P2X3	-	76 uM	-	-
TRPV1	~6 uM	~19 uM	-	~1 uM (AM404)
ASIC1a	Allosteric	-	-	-
Nav1.8	-	-	-	nM (AM404)
CB1	-	-	-	Indirect
PPARg	Weak	-	-	-

4. L2 - Pharmacokinetics

Pharmacokinetic profiles differ substantially. The most clinically relevant distinction is the disconnect between plasma half-life and tissue half-life - diclofenac exemplifies this with 1.2 h plasma t1/2 but 8-12 h synovial duration. Paracetamol has uniquely low protein binding (20%) and a non-linear toxicity pathway (NAPQI/glutathione depletion). Celecoxib is CYP2C9-dependent, creating a pharmacogenomic vulnerability.

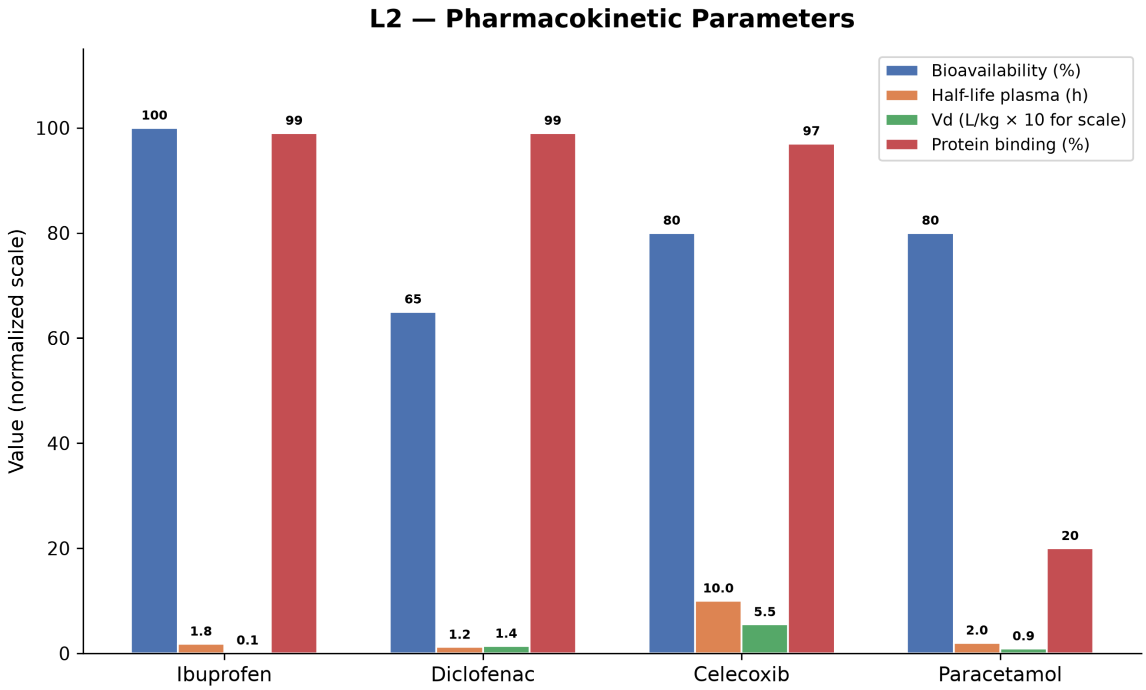

Figure 3. Key PK parameters across the four PoC drugs. Vd values shown x10 for scale visibility.

Parameter	Ibuprofen	Diclofenac	Celecoxib	Paracetamol
Bioavail.	100%	65%	80%	80%
Vd (L/kg)	0.1	1.4	5-6	0.9
Protein bind.	99%	99%	97%	20%
t1/2 plasma	1.8 h	1.2 h	8-12 h	2.0 h
t1/2 synovial	4-5 h	8-12 h	~12 h	N/A
Metabolism	Gluc+CYP2C9	CYP2C9+bil.	CYP2C9 only	Gluc+sulf.
Active metab.	No	Minor	No	AM404
Dosing	QID-TID	BID	QD-BID	QID

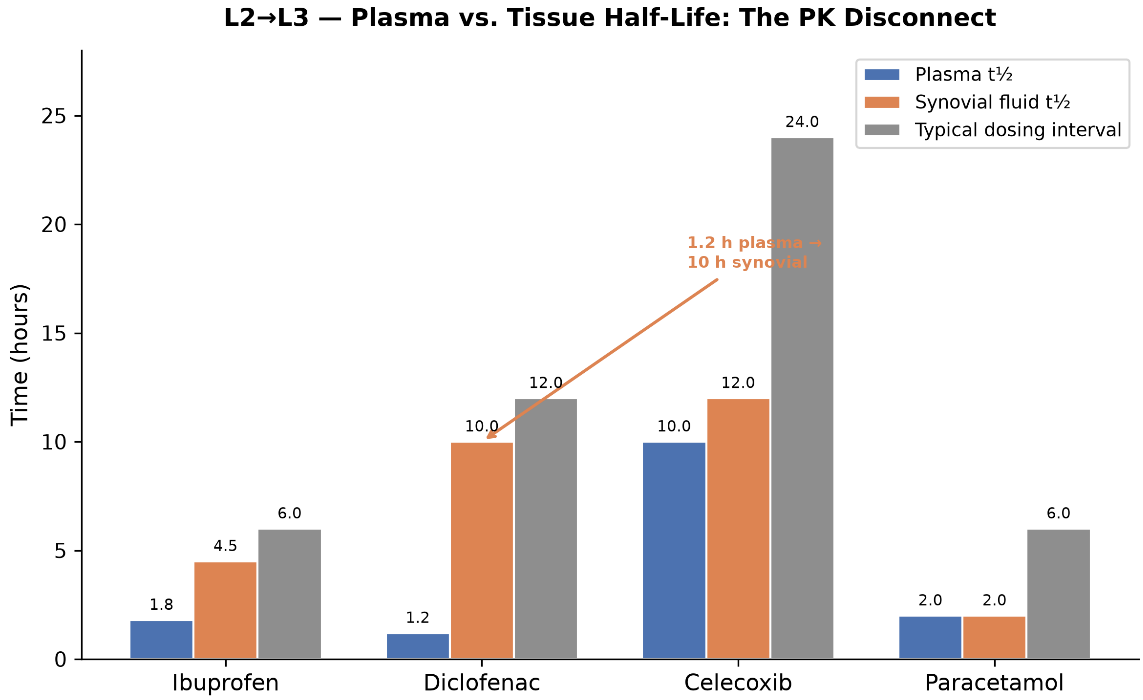

Figure 4. Plasma vs. synovial fluid half-life vs. dosing interval. Diclofenac shows the most dramatic disconnect: 1.2 h plasma but 8-12 h synovial t1/2, explaining BID dosing.

5. L3 - Systems Response

The systems level captures the biological consequences of L1 binding at the tissue and pathway level. This is the most novel and least standardized level in the framework - no structured database exists for systems-level drug effects, so L3 is populated through literature mining. An important caveat: several L3 dimensions in Figure 5 are labelled L1->L3 predicted because they are direct consequences of binding selectivity, not independent empirical findings at the tissue level. These are shaded in yellow in the heatmap. Independent L3 findings (in green) represent tissue-level data (e.g., synovial fluid residence time) not derivable from binding affinity alone. Where evidence is LOW or VERY LOW (single study, in vitro only), cells are marked with a diamond (asp) symbol.

Key L3 findings across the four PoC drugs, with circularity explicitly identified:

- Predicted (L1->L3): Diclofenac has the most potent dual COX inhibition, celecoxib's COX-2 selectivity predicts GI-sparing and pro-thrombotic effects, paracetamol has negligible COX effect at therapeutic doses.

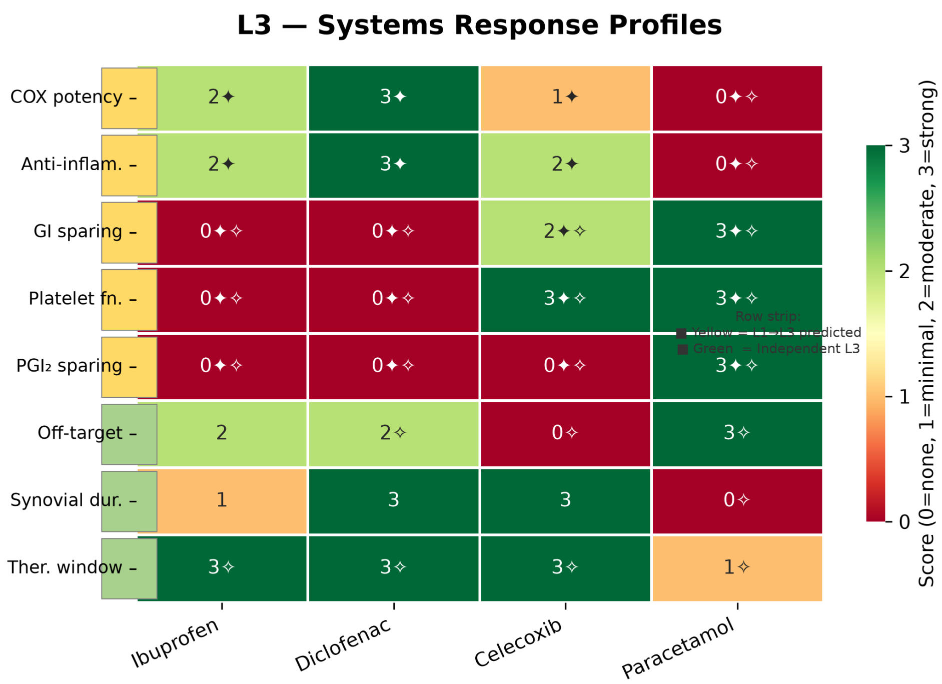

Figure 5. Systems response profiles annotated with circularity (row strip: yellow = L1->L3 predicted, green = independent) and evidence confidence (diamond = LOW/VERY LOW). Scores: 0 = absent, 1 = minimal, 2 = moderate, 3 = strong.

6. L4 - Clinical Outcomes

NNT for >=50% pain relief over 4-6 hours vs. placebo, with dose-specific values. Safety outcomes stratified by event type.

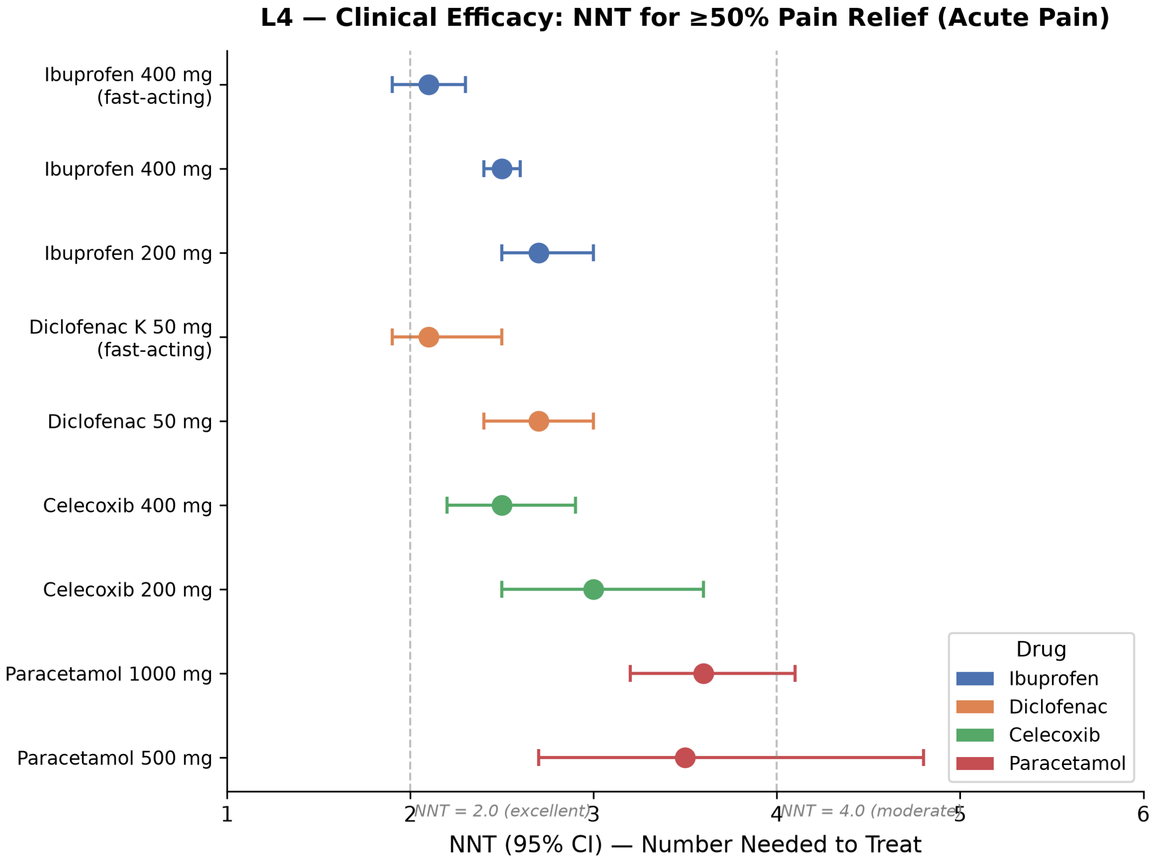

Figure 6. Forest plot of NNT values with 95% CI. Dashed lines at NNT=2.0 (excellent) and NNT=4.0 (moderate).

Drug/Dose	NNT (95% CI)	Success Rate
Ibuprofen 400 mg FA	2.1 (1.9-2.3)	65%
Ibuprofen 400 mg	2.5 (2.4-2.6)	54%
Ibuprofen 200 mg	2.7 (2.5-3.0)	46%
Diclofenac K 50 mg FA	2.1 (1.9-2.5)	~64%
Diclofenac 50 mg	2.7 (2.4-3.0)	~55%
Celecoxib 400 mg	2.5 (2.2-2.9)	~50%
Celecoxib 200 mg	3.0 (2.5-3.6)	~40%
Paracetamol 1000 mg	3.6 (3.2-4.1)	46%
Paracetamol 500 mg	3.5 (2.7-4.8)	32%

Safety Summary

Safety Domain	Ibuprofen	Diclofenac	Celecoxib	Paracetamol
GI bleed risk	Moderate	Highest	Lowest	None
CV risk (chronic)	Moderate	Highest	High	None
Hepatotoxicity	No	Rare	No	Yes (NAPQI)
Renal risk	Yes	Yes	Yes	Minimal
Platelet function	Temporary down	Temporary down	Normal	Normal

7. Clinical Decision Scenarios

The following scenarios demonstrate why a multi-axis framework outperforms a single-score comparison. Each "best choice" is derived from the full 4-level profile, not from NNT alone.

Scenario	Best Choice	Why	Ref
Healthy, dental pain	Ibuprofen 400 mg	Best NNT, fast onset	[1]
Elderly OA + CV risk	Paracetamol	No CV risk	[2]
Elderly OA + GI risk	Celecoxib + PPI	COX-2 spares GI	[3]
Severe acute pain IM	Diclofenac IM	Potent COX + P2X3	[4]
Patient on warfarin	Paracetamol	No platelet/GI effect	[2]
Renal colic	Diclofenac IM	IM + ureteric P2X3	[4]
Post-op multimodal	Paracet+ibuprofen	Synergy NNT <2.0	[5]

[1] Cochrane 2009 PMID:19821340 [2] Mallet 2023 PMID:37016715 [3] SCOT trial PMID:39660078
[4] PMID:39763427 [5] Moore 2015 PMID:26544675

> Key insight: No single score can capture these trade-offs. Paracetamol has the worst NNT but is the safest choice for CV-risk patients. Diclofenac has the best IM efficacy but the highest CV risk. Celecoxib selectivity is simultaneously its advantage (GI) and disadvantage (CV) - the same molecular feature at L1 produces opposite effects at L4.

8. Framework Takeaways

What the 4 Levels Reveal (that a single score cannot)

Diclofenac vs ibuprofen for acute pain

Same NNT ballpark (2.5 vs 2.7), but diclofenac has P2X3 activity (unique), enterohepatic recirculation (unique), and higher CV risk. A single score can't tell you which to choose for which patient.

Celecoxib's paradox

NNT 2.5 - same as ibuprofen. Safer for GI (L4) driven by COX-2 selectivity (L1) -> GI COX-1 sparing (L3). Higher CV risk (L4) driven by the same feature -> PGI2 suppression (L3). The mechanism of both advantage and harm is the same molecular feature.

Paracetamol's incommensurability

NNT 3.6 looks worse than ibuprofen 2.5. But paracetamol and ibuprofen don't share a mechanism, don't share a risk profile, and the right choice depends on the patient.

Ibuprofen's hidden feature

Best NNT (2.5) plus ASIC1a activity - invisible to a COX-focused single assay.

Framework Capabilities

Capability	Example
Captures different mechanisms	AM404 vs P2X3 vs ASIC1a
Traces L1->L3->L4 causality	COX-2 selectivity -> PGI2/TXA2 -> CV events
Handles PK/L3 disconnect	Diclofenac 1.2h plasma -> 8-12h synovial
Stratifies by patient	Same drug, different patient -> different ranking
Accommodates prodrugs	Paracetamol->AM404, sulindac, nabumetone
Evidence-graded	PMID, evidence level, confidence per datum

Profile, not score: DQF does not rank drugs. It fingerprints them. A score says "drug X is Y." A fingerprint says "here is what drug X does at each level - you decide what matters for your patient." This is the philosophical core of the framework: the question determines the weight, not the framework. For a patient with GI risk, celecoxib ranks first. For a patient with CV risk, the same drug is last. DQF preserves both truths instead of collapsing them into one.

9. Limitations

The following limitations are structural - they reflect design choices and scope boundaries:

1. Domain restriction: PoC covers one drug class (NSAIDs/analgesics) with 4 drugs. Generalizability to other classes is untested.
2. Selection bias: All four drugs are among the most-studied in pharmacology. Drugs with sparse data may produce incomplete profiles.
3. L3 is the weakest level: No structured database exists for systems response. L3 is populated through literature mining - labor-intensive and subject to publication bias.
4. Binding affinity variability: Ki values vary by assay conditions. Precision implied by a single number is misleading.
5. Levels are not independent: L1->L3->L4 forms a causal chain. Information is double-counted by design.
6. Emerging evidence: Several findings from 2025-2026 lack independent replication. Tagged as such in profiles.
7. No clinical validation: Claims of improved clinical decision-making are conceptual. No user study performed.
8. The framework does not rank: It profiles. Rankings require context-specific weights only a user can supply.
9. RAG reproducibility: The PubMed RAG endpoint is third-party hosted. Raw outputs were not archived at query time.

10. Methods & Data Sources

RAG System

Component	Detail
Endpoint	balade-pubmed-rag-bot.hf.space
Index	27.7M PubMed abstracts (1975 - Jan 2026)
Embedding	bge-small-en-v1.5 (FAISS IVF-PQ)
Reranker	cross-encoder MiniLM-L-6 (k=3)

Structured Data Sources

Level	Primary Source	Type
L1 Binding	PDSP Ki database + literature	Ki values, PMID-tagged
L2 PK	Inxight FRDB, DrugBank, Lombardo	ADME parameters
L3 Systems	PubMed RAG (primary)	Pathway/tissue data
L4 Clinical	Cochrane reviews, Oxford League	NNT/NNH with CIs

Evidence Hierarchy

Level	Definition	Example
HIGH	Multiple consistent studies, meta-analyses	Cochrane reviews
MODERATE	Replicated findings, some heterogeneity	Binding SAR studies
LOW	Single study, in vitro only	Emerging mechanisms
VERY LOW	Expert opinion, extrapolated	Computational predictions

Conclusion: The 4-level framework is viable. The most distinctive contribution is the L1 off-target profile + L3 systems response - these levels differentiate drugs within a class where NNT alone cannot. The framework does not rank drugs. It fingerprints them. The user asks "for this patient, with this condition, what matters?" - not "which drug is best."